

23/07/2020
Wednesday

Test 3

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Name : Trupti Vasant Gaikwad.

* 1. for loop (3 que)

Q. 1 write a program to print the multiplication table of given no using for loop.

→ `n = int(5)` (1)
`for i in range(1, 11):`
`Print(n, "*", i, "=", n*i)`

2) write a program to print all prime numbers between 1 & 100 using for loop. (1)

→ `n = int("enter a number:")` for num in range(2, 101)
Prime True.
`for i in range(2, num):`
`if i % num == 0:`
Prime False
`break`
`Print(num)`

3) write a program to find the sum of all even numbers from 1 to n using for loop.

→ `n = int("enter a number:")`
`sum = 0`
`for i in range(2, n+1, 2):` (1)
`sum += i`
`Print(sum)`

* 2) while loop (3 que)

Q. 1 write Program to calculate sum of digit number

→ `num = int("enter a number:")`

`sum = 0`

`while num > 0:`

`digit = num % 10`

`sum += digit`

`num = num // 10`

`Print(sum)` (1)

4) OOPS (Theory - 9 questions)

1) What is OOP?

→ OOPS is a programming paradigm based on class, object, inheritance etc. (1)

2) explain difference class & object with eg.

→ class: class is a Blueprint (template) it is used to create any object.

object: Instance of class. (6)

eg: class Student # (class name) (1)

Pass

s1 = student() # (object)

3) what are four Pillars of OOP? explain each

→ 1) Inheritance ✓

2) Polymorphism ✓

3) encapsulation ✓

4) Data abstraction ✓ (1)

4) what is inheritance? explain its types.

→ Inheritance: Inheritance allow one class Properties inherit another class.

Types 1) Single / Simple → one parent one child

2) Multiple → many parent one child

3) Multilevel → level by level

4) Hierarchical^{hi} → one parent many child

5) Hybrid → combination Multiple + Multilevel & Hierarchical + Multiple *

(1)

5) what is Polymorphism? explain Method overloading & overriding.

→ Polymorphism: ~~one~~ ^{many} class / forms ~~one~~ interface

1) Method overloading → Same method name & different Parameter (diff. Signature)

2) Method overriding → same signature (same method name & same Parameter)

6) what is encapsulation? what are its advantages?

→ encapsulation: wrapping of data / data hiding

Advantage: 1) Data Security
2) data hiding

7) what is Abstraction? How is it implemented in Python?

→ Abstraction hides implementation modules.
Module: from ABC ^{import} abc, abstractmethod

8) what is the difference between Abstraction & encapsulation?

→ Abstraction: Hides implementation modules.
encapsulation: Hides data

9) explain the use of super() function in Python.

→ It is used to call parent class constructor from child class.

*3) Functions (5 question):

5) difference between user-defined & built-in.

→ user-defined function: created by user.

I am Satisfied!

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built in function: Already created. Built-in function

4) what is function? explain advantage.

→ function: function is block of statement that perform specific task.
advantage: 1) code reusability. (1)

1) write function to calculate factorial of number.

→ def factorial(n):

fact = 1

for i in range(1, n+1):

fact *= i

return fact

num = int(25)

Print(factorial(num)).

*2) while Loop: (3 que):

1) write Program reverse no using while loop.

→ num = int(42215)

rev = 0

while num > 0:

digit = num % 10

rev = rev * 10 + digit

num = num // 10

Print(rev).

3) Function (3 que):

2) write function that takes list input return largest element:

→ while

def largest(list):

return max(list)

list1 = [10, 20, 40, 50, 5, 60]

Print(list1)